

Autodesk® Scaleform®

Scaleform "A

⌘ ⌘ "A DirectX 9 ♠ ⌘ Scaleform ⌘ 3D ⌘ ⌘ ⌘ .

: Ben Mowery
♣: 3.04
√+: : 2011 4 8 4 ■

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[\(http://gameware.autodesk.com/scaleform/developer/\)](http://gameware.autodesk.com/scaleform/developer/) / "A ■ . Q ‡ ♪
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libjpeg ↑ zlib Scaleform

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■ AE ³ ö AEScaleform AEGFx SDK 4.2 AEGFx Players AEGFx Player D3D9.
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      ■  .  f  "A  "      "      Q'  £      ‡  £  -  Ç
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```

Scaleform Unreal® Engine 3, Gamebryo™, Bigworld®, Hero Engine™, Touchdown Jupiter Engine™, CryENGINE™ Trinigty Vision Engine™

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DXUT ♣ İ || / 7 π Я 4 Ω M 7 π "E Q' /
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[Gfx::Loader](#) - √ || 7 π █ ♣ Y || ↑ 3 [Gfx::MovieDef](#)
 . 8 ♠ [Render::Renderer2D](#) - √ [Render::HAL](#) - √ ||

Scaleform ~ API (/ é DirectX) Q' ♣ / "A || ♣ π f ."
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♣ Q' ♫ ShadowVolume ♣ Tutorial\Section4.1 / "A 6i █ . ♣
 √ "A / "A 7 ï ♣ "E ï 6i .

■ ShadowVolume.cpp / 4 .

```
#include "GFx_Kernel.h"
#include "GFx.h"
#include "GFx_Renderer_D3D9.h"
```

```
GFxTutorial * g_pGFxTutorial = 0;

// One GFx::Loader per application
Loader g_loader;

// One GFx::MovieDef per SWF/GFx file
Ptr<MovieDef> pMovieDef;

// One GFx::Movie per playing instance of movie
Ptr<Movie> pMovie;

// Renderer data
Ptr<Render::D3D9::HAL> pRenderHAL;
Ptr<Render::Renderer2D> pRenderer;
MovieDisplayHandle hMovieDisplay;
```

```
GFx::System * g_pGFxSystem = 0;

// One GFx::System per application
GFx::System g_system;
```

```
GFx::System * g_pGFxSystem = 0;

// One GFx::System per application
GFx::System g_system;
```

```
Gfx::System    ÿ    ‡    ĭ ‡ ‡ ¨    ♣ /    / "A    ‡ † -    .
f    "A,    ÿ    ♣    ■    '    .    ‡    €    GfxTutorial ‡    ‡    ♣ /
    ‡ j    1 ‡    .
```

```
    ‡    ¨    ‡    π ^    w /    f    .    r , Gfx::System::Init()    Gfx::System::Destroy()
♠ ♣    ^    1 ‡ Gfx::System    _    ‡    r    ÿ    1    1 ‡    .
```

```
† †    Scaleform    1    InitApp()    ‡ / "A    ‡    ĭ ‡    π    WinMain ‡    □    ♠
1    ■    ‡ q    .    ^    ŵ    -    ÿ    InitApp() /    1
```

```
gfx = new GfxTutorial();
assert(gfx != NULL);
if(!gfx->InitGfx())
    assert(0);
```

```
GfxTutorial    Gfx::Loader    _    _    ~    ■    .    ■    ♠    ‡    ĭ ‡    † π
Gfx::Loader    _    _    ~    ■    € ‡    _    _    SWF/Gfx    ħ    π    bi'    ħ    π
□    ‡    /    p    ,    € ‡    w    /    □    ^    Q'
■    =    .    π    SWF/Gfx    ■    ‡ Ç    -    □    ^    "Q    j    p
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ē ♠    Ũ'
```

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GfxTutorial::InitGfx() / "A    _    '    Gfx::Loader /    Ũ'    ē ♠    1 ‡    .
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.    ‡    /    Scaleform    PlayerLog    θ    Q'    ,    ‡    θ
j    /    ^    .    r    =    ■    π    Ÿ    '    π
Gfx::Log    Ũ'    ^    "A    r    θ    ■    1 ‡    .
```

```
// Initialize logging -- Scaleform will print errors to the log
// stream.
gfxLoader->SetLog(Ptr<Log>(*new Log()));
```

```
Gfx::Loader    Gfx::FileOpener    ħ    π    ■    .    ^    /    ē
Ÿ    ■    ■    r    j    p    "    □    ■    π    Q'    □    bi
Gfx::FileOpener    Ũ'    3    .
```

```
// Give the loader the default file opener
```



```

    Gfx::MovieDef _ V Gfx::Movie
Gfx::Movie
    pUIMovie = *pUIMovieDef->CreateInstance(true, 0, NULL);
    assert(pUIMovie.GetPtr() != NULL);

```

```

CreateInstance
false
    i
    Memory System Overview

```

```

    Advance()
false CreateInstance

```

```

// Advance the movie to the first frame
pUIMovie->Advance(0.0f, 0, true);

// Note the time to determine the amount of time elapsed between
// this frame and the next
MovieLastTime = timeGetTime();

```

```

Advance
    1
    1
    1

```

```

3D
pUIMovie->SetBackgroundAlpha(0.0f);

```

```

    3D
    3D

```

```

    DirectX
    Render::D3D::HALInitParams

```

```

Render::D3D9::HAL::InitHAL(
    Scaleform
    InitHAL
    ShadowVolume
    OnResetDevice
    OnResetDevice

    pRenderHAL->InitHAL(
        Render::D3D9::HALInitParams(pd3dDevice, presentParams,
            Render::D3D9::HALConfig_NoSceneCalls));

    InitHAL
    HAL_NoSceneCalls
    DirectX BeginScene() EndScene()
    ShadowVolume
    OnFrameRenderer

    w
    1
    ShadowVolume
    OnLostDevice
    GfxTutorial
    OnLostDevice

    pRendererHAL->ShutdownHAL();

    DXUT
    Win32/DirectX
    Scaleform SDK
    GfxPlayerTiny.cpp

    Scaleform
    GfxTutorial
    WinMain

    delete gfx;
    gfx = NULL;

```

DirectX 9 . Scaleform
 InitHAL()
 ShutdownHAL()

DirectX9 / "A InitHAL D3DPOOL_DEFAULT
 InitHAL

ShutdownHAL D3DPOOL_DEFAULT . ShadowVolume "A
 DXUT OnResetDevice / "A
 D3DPOOL_DEFAULT OnLostDevice / "A
 GfxTutorial::OnLostDevice ShutdownHAL GfxTutorial::OnResetDevice
 InitHAL

D3DPOOL_DEFAULT
 InitHAL ShutdownHAL

GFxTutorial::OnResetDevice /
 GFxTutorial::OnResetDevice /

```
// Use the window client rect size as the viewport.
RECT windowRect = DXUTGetWindowClientRect();
DWORD windowWidth = windowRect.right - windowRect.left;
DWORD windowHeight = windowRect.bottom - windowRect.top;
pUIMovie->SetViewport(windowWidth, windowHeight, 0, 0,
    windowWidth, windowHeight);
```

[SetViewport](#) PC
 Scaleform

OpenGL
 OpenGL

Scaleform
'
'

ShadowVolume OnFrameRender()
BeginScene() EndScene()
GfXTutorial::AdvanceAndRender()

```
void AdvanceAndRender(void)
{
    DWORD mtime = timeGetTime();
    float deltaTime = ((float)(mtime - MovieLastTime)) / 1000.0f;
    MovieLastTime = mtime;

    pUIMovie->Advance(deltaTime, 0);
    pRenderer->BeginFrame();

    if (hMovieDisplay.NextCapture(pRenderer->GetContextNotify()))
    {
        pRenderer->Display(hMovieDisplay);
    }

    pRenderer->EndFrame();
}
```

Advance deltaTime
GfX::Movie::Advance /

[Render::Renderer2D::Display](#) D3D
DirectX
Render::Renderer2D::Display
Display
Scaleform
DX9

```

DX9  Ů      ‡      Ĩ ‡ ‡ —  X ‡ ‡  'O r      e
GfxTutorial::AdvanceAndRender()  ʌ      Æ /  Q'      .

```

```

// Save DirectX state before calling Scaleform
g_pStateBlock->Capture();
// Render the frame and advance the time counter
gfx->AdvanceAndRender();
// Restore DirectX state to avoid disturbing game render state
g_pStateBlock->Apply();

```

```

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‡ʌ      ‡♣      Ÿ      ˘      . DXUT      '      ° ö      ‡      □.

```

```

// Disable default UI

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Scaleform      π UI      ■      ShadowVolume ‡      Ĩ ‡

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Gfx::Movie::SetViewport Λ π $\mathfrak{z}$ $\mathfrak{U}$ $\blacksquare$ $=$ $\cdot$ r $-$ $\mathfrak{z}$
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$\neq$ $4:3$ $\neq$ $\mathfrak{H}$ $\mathfrak{z}$ $\mathfrak{p}$ $/$ $"A$ $\check{\alpha}$ Λ $\blacksquare$ $\neq$ $\mathfrak{U}$ $\cdot$
Scaleform π $\mathfrak{J}$ $\mathfrak{X}$ $\uparrow$ $\neq$ $"A$ $\neq$ $\mathfrak{U}$ $\mathfrak{z}$ $\mathfrak{U}$
 $\square$ $=$ $\neq$ $\mathfrak{J}$ $\neq$ $\cdot$ $\vdash$ π $\mathfrak{t}$ $/$ $\acute{\epsilon}$ $\blacksquare$ $\spadesuit$ $\text{---}\%$ $\square$ θ $\neq$ $"A$
 $\mathfrak{z}$ $\mathfrak{U}$ $/$ f π $\mathfrak{t}$ $\text{---}"E$ $-$ $($ $, 800 \times 600$ π $\mathfrak{U}$ $\mathfrak{Y}$ $\neq$
 1600×1200 π $\sim$ $\mathfrak{U}$ $\mathfrak{Y}$ $\neq$). Scaleform $\blacksquare$ $\check{\alpha}$ Λ $\mathfrak{p}$ $\ddot{o}$ $\mathfrak{U}$ $\blacksquare$
 $\cdot$ $"$ $,$ $\vdash$ π $\mathfrak{t}$ $\neq$ $\spadesuit$ $\heartsuit$ $\leftarrow$ $"A$ $\mathfrak{t}$ $\cdot$

[Gfx::Movie::SetViewScaleMode](#) w $\heartsuit$ $\neq$ $\neq^{\vee}$ $=$ $\neq$ $\mathfrak{J}$ π $\cdot$ π $\mathfrak{z}$ $/$
 $\mathfrak{L}$ Ω $/$ $\mathfrak{J}$ $\mathfrak{d}$ Λ $'$ $-$ Λ GfxTutorial::InitGfx() π
 i $/$ Λ $\sim$ Gfx::Movie $_$ $\mathfrak{V}$ $\mathfrak{E}$ $\mathfrak{J}$ $\mathfrak{E}$ $\mathfrak{J}$ Λ $\blacksquare$ $\cdot$

pUIMovie->SetViewScaleMode(Movie::SM_ShowAll);

$\neq$ $\mathfrak{U}$ $"A$ $/$ $\mathfrak{z}$ SetViewScaleMode $/$ $'$ Λ $\mathfrak{E}$ $\mathfrak{J}$ $\neq$ $\mathfrak{q}$ $\mathfrak{O}$ $'$ $-$

SM_NoScale	$\check{\alpha}$ Λ π $\neq$ π $\mathfrak{U}$ $\sim$ $\mathfrak{J}$ $\cdot$
SM_ShowAll	π $\mathfrak{z}$ $\mathfrak{z}$ $"A$ $/$ $\mathfrak{d}$ Λ $\check{\alpha}$ Λ w $\heartsuit$ $\cdot$
SM_ExactFit	$\mathfrak{z}$ $/$ $\mathfrak{U}$ $'$ $\neq$ $\clubsuit$ $\mathfrak{V}$ $\check{\alpha}$ Λ w $\heartsuit$ $\cdot$ $\cdot$ r $\cdot$ $\neq$ $\mathfrak{Y}$ $\blacksquare$ $\cdot$
SM_NoBorder	$\mathfrak{z}$ $\mathfrak{z}$ $"A$ $\clubsuit$ $\mathfrak{V}$ $\check{\alpha}$ Λ w $\heartsuit$ $\cdot$ $\cdot$ r $\neq$ $\mathfrak{Y}$ $\blacksquare$ $\cdot$

Λ $\neq$ [SetViewAlignment](#) γ $\check{\alpha}$ Λ π Λ $\mathfrak{L}$ $\neq$ $\cdot$ $\mathfrak{z}$ $\neq$
 $\mathfrak{U}$ SM_NoScale $"$ SM_ShowAll $\acute{\epsilon}$ $\leftarrow$ $\mathfrak{X}$ $\blacksquare$ $\neq$ $\blacksquare$ $\cdot$

SetViewAlignment ☉ / "A ǎ ʌ ‡ Ÿ / й ' ʃ . ‡ € † ‡

 ʒ ʋ ʌ 𐀀 / й 𐀀 - .

```
pUIMovie->SetViewAlignment(Movie::Align_CenterRight);
```

W♥ 3 ≠ ı ≠ "E ≠ ≠ " =

SetViewScaleMode SetViewAlignment π 3 ← □ .

SetViewAlignment SetViewScaleMode SM_NoScale > ⚙ ↶ ↷
⌂ ~ ± 🔍 ↳ Ω . ○ 🎵 , w♥, Nw🎵 Ÿ π ~ Q' /
Scaleform ⌋ □ π N é 🖼️ ▶
⋮ Y d3d9guide.fla / 🖼️ .

`Stage` `(Stage.scaleMode, Stage.align)` `. SetViewScaleMode`

Scaleform $\int \pi \tilde{U} f \Omega \sim , \mathcal{B} \pi ! \wedge \tilde{U} / \gamma \tilde{U} !$
 $\cdot \quad \neq - \odot = \gamma \cdot$

#1: $\exists$ 

```
pMovie->SetViewport(screen_width, screen_height, 0, 0,
                    screen_width, screen_height, 0);
```

$\neq \leftarrow \neq ' \cdot \pi ! \leftarrow (0, 0) / \wedge \sim \text{■} \mathfrak{U}^u /$
 $\int \pi ! \wedge \tilde{U} / \tilde{U} ! \cdot \odot \vee "A \int \tilde{U}$
 $! \quad w \heartsuit \cdot$

#2: $\exists$  $r, \exists \pi ! \wedge \tilde{U} / \wedge \cdot$

```
pMovie->SetViewport(screen_width, screen_height, 0, 0,
                    screen_width / 4, screen_height / 4, 0);
```

$\neq \leftarrow / \neq ' \cdot \pi \text{■} \tilde{\omega} \square \mathfrak{U}^u / \pi \wedge \cdot$
 $\uparrow \text{HandleEvent} \quad \exists ! \mathfrak{D} \pi ! \wedge \tilde{U} / \tilde{U} ! \neq \mathfrak{b} \neq$
 $, \cdot ! ' \int \tilde{U} / \pi "A \odot \spadesuit \vee \cdot$

#3: $\exists$  $/ \wedge$

```
movie_width = screen_width / 6;
movie_height = screen_height / 6;
pMovie->SetViewport( screen_width, screen_height,
                    screen_width / 2 - movie_width / 2,
                    screen_height / 2 - movie_height / 2,
                    movie_width, movie_height);
```

$\neq \leftarrow \exists ! \pi \neq \neq \cdot \int \neq \tilde{U} (0, 0) / ' \mathfrak{b} (\text{screen_width} /$
 $2 - \text{movie_width} / 2, \text{screen_height} / 2 - \text{movie_height} / 2) / \text{■} \mathfrak{U} \wedge / =$
 $\clubsuit ! > / "A \tilde{A} - \cdot$

$\tilde{\alpha} \wedge \mathcal{K} / \text{■} \sim \uparrow \text{Gfx::Movie::SetViewAlignment} / \pi "A \mathcal{B} \neq$
 $\text{■} \exists \neq \neq ' \int / \pi \cdot \mathfrak{L} \neq \text{Gfx::Movie::SetViewport} ! ' /$
 $! \quad \tilde{U} \spadesuit \neq \leftarrow \exists \text{SetViewAlignment} \text{SetViewScaleMode} \mathcal{B} '$
 $w \heartsuit \neq \text{Scaleform} \odot \spadesuit \vee$

й $\neq$ [Gfx::Movie::HandleEvent](#) ч [. Gfx::KeyEvent](#) Gfx::CharEvent
 х $\neq$  .

```
KeyEvent(EventType eventType = None,
          Key::Code code = Key::None,
          UByte asciiCode = 0,
          UInt32 wcharCode = 0,
          UInt8 keyboardIndex = 0)
```

```
CharEvent(UInt32 wcharCode, UInt8 keyboardIndex = 0)
```

Gfx::KeyEvent $\leftarrow$ й $\text{Q} \sim$, Gfx::CharEvent ч ASCII $\text{v} \square$ Q .
 / "A Gfx::CharEvent WM_CHAR г / $\tilde{\text{Y}} \text{E}$. Gfx::KeyEvent
 WM_SYSKEYDOWN, WM_SYSKEYUP, WM_KEYDOWN WM_KEYUP г /
 $\tilde{\text{Y}} \text{E}$. ц o 1 $\neq$.

y SHIFT y $\tilde{\text{U}}$ / "A 'c' y : WM_KEYDOWN г /
 Gfx::KeyEvent $\tilde{\text{Y}} \text{E}$ $\neq -$ e ' - O Q л 1 $\neq$.
 - 'c' $\sim$, π $\leftarrow$ й
 - y $\sim$  . $\sim$
 - SHIFT  х $\neq$.

y $\neq$ / " ч q" ASCII > 'C' Scaleform $\clubsuit$ WM_CHAR /
 Gfx::CharEvent $\neq -$.

y 'c' й $\leftarrow$ WM_KEYUP г / Gfx::KeyEvent $\text{O} -$.
 й $\leftarrow$ Gfx::CharEvent E ' .

y F5 y : y $\sim$ Gfx::KeyEvent $\text{O} \text{e}$ $\sim$ $\neq$
 E . F5 л ASCII й / O $\sim \text{v}$ / Gfx::CharEvent E
 ' .

π Gfx::KeyEvent $\neq$ y $\sim$ ' $\neq$ $\clubsuit$. / $\spadesuit$
 , й Gfx_Event.h  / "A л π O $\spadesuit$ Q
 й л . GfxPlayerTiny.cpp Scaleform Player a $\leftarrow$ й

Scaleform ♠ ♣
 ■ ♠ ♣ M J D Q / ■ . ♣
 , ^ ♣ , V' TTF ♣ Ÿ |
 ■ . π 1 ŵ ŵ π J D Q ς .
 ♣ ♣ ♣ / "A Ÿ = ■ .
 Scaleform π ~ ♣ / ♣ V u "A , □ u "A ♣ / ↑ ■ ♣
 ♣ <http://gameware.autodesk.com/scaleform/developer/?action=doc> ♣ .
 ♣ ♠ ♣ ♣ V π □ V ~ ♣ □
 V 3 V ♣ Q . ♣ "A
 ■ ♣ = ♣ / 1 V . u □ J D Q ♣
 7 ♣ r 5000 , π ♣ ♣ ŵ ŵ π ♣ ♣ . u , π X
 V J D ♣ "A
 ■ / - u ♣ .
 Scaleform ♠ ~ ♣ u ♣ V ♣ ■ ♣ ~ / u □ ♣
 ~ ♣ ~ V . "A ■ Scaleform π
 (glyph-packing) Q~ ■ ~ ~ / "A ~ . Scaleform
 —% , ♣ ♣ ε π u □ / "A ↑ π TTF J D / P D 3
 . ♣ . i Scaleform "fake italic" "fake bold" ♣ , ♣ 1 ♣ ♣
 J D ♥ - 3 "A π ♣ ♣ ♣ ♣ V ■
 ♣ ù ♣ ♣ ♣ . ♣ u Ω
 ♣ ♣ = ■ π ♣ 3 Ÿ' - / "A .
 Bin\Data\AS2\Samples\FontConfig / "A Ë ♣ ♣ sample.swf ■ Scaleform Player
 D3D9 / ♣ "A ♣ Ÿ □ .

4c 4d / "A, Scaleform "E ' E ‡ "A □ π 'O ~ , ‡ Ù

t □ .

, ' . ' Ÿ"E 3

∅ π й / ♠ ♥ è º . 1 ♠ ‡ "E Q' è º , /

↑ ■ . Ł ■ π ‡ gfx_2.1_texteffects_sample.zip ‡ q , □ / "A й

■ .

...
 Scaleform
 C++
 C++
 Scaleform
 C++

Scaleform C++
 C++
 FSCOMMAND
 flash.external.ExternalInterface.
 (6.1.2 / "A").
 ExternalInterface
 fscommand
 -fstree, -fslst
 -fspams
 SWF
 ExternalInterface / "A"
 fscommand

ActionScript
 fscommand
 "setMode", "2"

fscommand("setMode", "2");

```

    > fscommand / "A"
    . ExternalInterface

```

```

    Gfx FSCCommand
    Gfx::FSCCommandHandler
    Gfx::Movie
    fscommand
    GfxPlayerTiny
    "E
    ("FxPlayerFSCCommandHandler"
    1) ShadowVolume /
    Tutorial\Section6.1 /

```

```

    Gfx::FSCCommandHandler
    "A

```

```

class OurFSCCommandHandler : public FSCCommandHandler
{
public:
    virtual void Callback(Movie* pmovie,
                          const char* pcommand, const char* parg)
    {
        GfxPrintf("FSCCommand: %s, Args: %s", pcommand, parg);
    }
};

```

```

    fscommand
    fscommand

```

```

    GfxTutorial::InitGfx() / Gfx::Loader
    "E
    "

```

```

// Register our FSCCommand handler
Ptr<GfxFSCCommandHandler> pcommandHandler = *new OurFSCCommandHandler;
gfxLoader->SetFSCCommandHandler(pcommandHandler);

```

```

Gfx::Loader
    Gfx::Movie
    SetFSCCommandHandler
    "A

```

```

    fscommand
    "Toggle Fullscreen"
    . UI
    ShadowVolume

```

```

FSCCommand: ToggleFullscreen, Args:

```

ÿ / "A d3d9guide.swf ■ ~ B (F9) . 3 / 4
 ‡ Symbol Definition(s) Æhud Ævar : Frame 1:/ "A Ÿ"E fscommand() 4 .

fscommand() 4

Û ̂ fscommand("ToggleFullscreen") fscommand("ToggleFullscreen", this.UI_visible)
 "A this.UI_visible > C++ / 4 . f ■ : "A(CTRL+ALT+Shift+S)
 d3d9guide.swf 4 .

UI Ÿ' π FSCCommand ‡ Ÿ 3 ♠ ♥ ÿ — "Q ShadowVolume
 Y / ■ . "A, π ÿ fscommand 0 / 4 ♣ 4 3 ° ÿ
 4 .

```
if (strcmp(pcommand, "ToggleFullscreen") == 0)
    doToggleFullscreen = true;
```

DXUT ̄ OnFrameMove ■ ♣ / 4 . OnFrameMove 4 /
 ÿ 4 .

```
if (doToggleFullscreen)
{
    doToggleFullscreen = false;
    DXUTToggleFullScreen();
}
```

π ExternalInterface.call fscommand $\mathcal{O}$ $\uparrow$ ♣ $\parallel$ $\square$ $\vee$ $>$
 N $\mathfrak{z}$ $\mathfrak{z}$ u / $\dot{\epsilon}$.

```

class OurExternalInterfaceHandler : public ExternalInterface
{
public:
virtual void Callback(Movie* pmovieView,
                     const char* methodName,
                     const Value* args,
                     unsigned argCount)
{
    GfxPrintf("ExternalInterface: %s, %d args: ",
              methodName, argCount);
    for(unsigned i = 0; i < argCount; i++)
    {
        switch(args[i].GetType())
        {
            case Value::VT_Null:
                GfxPrintf("NULL");
                break;
            case Value::VT_Boolean:
                GfxPrintf("%s", args[i].GetBool() ? "true" : "false");
                break;
            case Value::VT_Number:
                GfxPrintf("%3.3f", args[i].GetNumber());
                break;
            case Value::VT_String:
                GfxPrintf("%s", args[i].GetString());
                break;
            default:
                GfxPrintf("unknown");
                break;
        }
        GfxPrintf("%s", (i == argCount - 1) ? "" : ", ");
    }
    GfxPrintf("\n");
}
};

```

```
Ptr<ExternalInterface> pEIHandler = *new OurExternalInterfaceHandler;  
gfxLoader.SetExternalInterface(pEIHandler);
```

Ÿ"E . F9 Ы ~ Symbol Definitions() Æhud Ævar : Frame 1 /

MeshPathFocus

ExternalInterface ∖ ExternalInterface ∅ — ~ , ⋮ Ũ □ (" ") π

ǎ Ũ ♣ e, ■ π π { ≠ ‡ "MeshPathFocus" ♣ . ⋮ ≠ "A

∃ π ^ ⋮ ≠ ~ ǎ £ √ .

∖ ' } Q 1 ≠ .

```
ExternalInterface: MeshPathFocus, 1 args: false
Focus change:
    _level0.hud.text_MeshPath.field => null
ExternalInterface: MeshPathFocus, 1 args: true
Focus change:
    null => _level0.hud.text_MeshPath.field
```

卩 Ө 𪖇 Ǟ 𐌿 𐌶 𐌸 𐌹 𐌺 𐌻 𐌼 𐌽 𐌾 𐌿 𐍂 𐍃 𐍄 𐍅 𐍆 𐍇 𐍈 𐍉 𐍊 𐍋 𐍌 𐍍 𐍎 𐍏 𐍐 𐍑 𐍒 𐍓 𐍔 𐍕 𐍖 𐍗 𐍘 𐍙 𐍚 𐍛 𐍜 𐍝 𐍞 𐍟 𐍠 𐍡 𐍢 𐍣 𐍤 𐍥 𐍦 𐍧 𐍨 𐍩 𐍪 𐍫 𐍬 𐍭 𐍮 𐍯 𐍰 𐍱 𐍲 𐍳 𐍴 𐍵 𐍶 𐍷 𐍸 𐍹 𐍺 𐍻 𐍼 𐍽 𐍾 𐍿 𐎀 𐎁 𐎂 𐎃 𐎄 𐎅 𐎆 𐎇 𐎈 𐎉 𐎊 𐎋 𐎌 𐎍 𐎎 𐎏 𐎐 𐎑 𐎒 𐎓 𐎔 𐎕 𐎖 𐎗 𐎘 𐎙 𐎚 𐎛 𐎜 𐎝 𐎞 𐎟 𐎠 𐎡 𐎢 𐎣 𐎤 𐎥 𐎦 𐎧 𐎨 𐎩 𐎪 𐎫 𐎬 𐎭 𐎮 𐎯 𐎰 𐎱 𐎲 𐎳 𐎴 𐎵 𐎶 𐎷 𐎸 𐎹 𐎺 𐎻 𐎼 𐎽 𐎾 𐎿 𐏀 𐏁 𐏂 𐏃 𐏄 𐏅 𐏆 𐏇 𐏈 𐏉 𐏊 𐏋 𐏌 𐏍 𐏎 𐏏 𐏐 𐏑 𐏒 𐏓 𐏔 𐏕 𐏖 𐏗 𐏘 𐏙 𐏚 𐏛 𐏜 𐏝 𐏞 𐏟 𐏠 𐏡 𐏢 𐏣 𐏤 𐏥 𐏦 𐏧 𐏨 𐏩 𐏪 𐏫 𐏬 𐏭 𐏮 𐏯 𐏰 𐏱 𐏲 𐏳 𐏴 𐏵 𐏶 𐏷 𐏸 𐏹 𐏺 𐏻 𐏼 𐏽 𐏾 𐏿 𐐀 𐐁 𐐂 𐐃 𐐄 𐐅 𐐆 𐐇 𐐈 𐐉 𐐊 𐐋 𐐌 𐐍 𐐎 𐐏 𐐐 𐐑 𐐒 𐐓 𐐔 𐐕 𐐖 𐐗 𐐘 𐐙 𐐚 𐐛 𐐜 𐐝 𐐞 𐐟 𐐠 𐐡 𐐢 𐐣 𐐤 𐐥 𐐦 𐐧 𐐨 𐐩 𐐪 𐐫 𐐬 𐐭 𐐮 𐐯 𐐰 𐐱 𐐲 𐐳 𐐴 𐐵 𐐶 𐐷 𐐸 𐐹 𐐺 𐐻 𐐼 𐐽 𐐾 𐐿 𐑀 𐑁 𐑂 𐑃 𐑄 𐑅 𐑆 𐑇 𐑈 𐑉 𐑊 𐑋 𐑌 𐑍 𐑎 𐑏 𐑐 𐑑 𐑒 𐑓 𐑔 𐑕 𐑖 𐑗 𐑘 𐑙 𐑚 𐑛 𐑜 𐑝 𐑞 𐑟 𐑠 𐑡 𐑢 𐑣 𐑤 𐑥 𐑦 𐑧 𐑨 𐑩 𐑪 𐑫 𐑬 𐑭 𐑮 𐑯 𐑰 𐑱 𐑲 𐑳 𐑴 𐑵 𐑶 𐑷 𐑸 𐑹 𐑺 𐑻 𐑼 𐑽 𐑾 𐑿 𐒀 𐒁 𐒂 𐒃 𐒄 𐒅 𐒆 𐒇 𐒈 𐒉 𐒊 𐒋 𐒌 𐒍 𐒎 𐒏 𐒐 𐒑 𐒒 𐒓 𐒔 𐒕 𐒖 𐒗 𐒘 𐒙 𐒚 𐒛 𐒜 𐒝 𐒞 𐒟 𐒠 𐒡 𐒢 𐒣 𐒤 𐒥 𐒦 𐒧 𐒨 𐒩 𐒪 𐒫 𐒬 𐒭 𐒮 𐒯 𐒰 𐒱 𐒲 𐒳 𐒴 𐒵 𐒶 𐒷 𐒸 𐒹 𐒺 𐒻 𐒼 𐒽 𐒾 𐒿 𐓀 𐓁 𐓂 𐓃 𐓄 𐓅 𐓆 𐓇 𐓈 𐓉 𐓊 𐓋 𐓌 𐓍 𐓎 𐓏 𐓐 𐓑 𐓒 𐓓 𐓔 𐓕 𐓖 𐓗 𐓘 𐓙 𐓚 𐓛 𐓜 𐓝 𐓞 𐓟 𐓠 𐓡 𐓢 𐓣 𐓤 𐓥 𐓦 𐓧 𐓨 𐓩 𐓪 𐓫 𐓬 𐓭 𐓮 𐓯 𐓰 𐓱 𐓲 𐓳 𐓴 𐓵 𐓶 𐓷 𐓸 𐓹 𐓺 𐓻 𐓼 𐓽 𐓾 𐓿 𐔀 𐔁 𐔂 𐔃 𐔄 𐔅 𐔆 𐔇 𐔈 𐔉 𐔊 𐔋 𐔌 𐔍 𐔎 𐔏 𐔐 𐔑 𐔒 𐔓 𐔔 𐔕 𐔖 𐔗 𐔘 𐔙 𐔚 𐔛 𐔜 𐔝 𐔞 𐔟 𐔠 𐔡 𐔢 𐔣 𐔤 𐔥 𐔦 𐔧 𐔨 𐔩 𐔪 𐔫 𐔬 𐔭 𐔮 𐔯 𐔰 𐔱 𐔲 𐔳 𐔴 𐔵 𐔶 𐔷 𐔸 𐔹 𐔺 𐔻 𐔼 𐔽 𐔾 𐔿 𐕀 𐕁 𐕂 𐕃 𐕄 𐕅 𐕆 𐕇 𐕈 𐕉 𐕊 𐕋 𐕌 𐕍 𐕎 𐕏 𐕐 𐕑 𐕒 𐕓 𐕔 𐕕 𐕖 𐕗 𐕘 𐕙 𐕚 𐕛 𐕜 𐕝 𐕞 𐕟 𐕠 𐕡 𐕢 𐕣 𐕤 𐕥 𐕦 𐕧 𐕨 𐕩 𐕪 𐕫 𐕬 𐕭 𐕮 𐕯 𐕰 𐕱 𐕲 𐕳 𐕴 𐕵 𐕶 𐕷 𐕸 𐕹 𐕺 𐕻 𐕼 𐕽

```
if (strcmp(methodName, "MeshPathFocus") == 0 && argCount == 1)
{
    if (args[0].GetType() == Value::VT_Boolean)
```

```

    gfx->SetTextboxFocus(args[0].GetBool());
}

```

```

GFXTutorial::ProcessEvent
{
    if (uMsg == WM_SYSKEYDOWN || uMsg == WM_SYSKEYUP ||
        uMsg == WM_KEYDOWN || uMsg == WM_KEYUP ||
        uMsg == WM_CHAR)
    {
        if (textboxHasFocus || wParam == 32 || wParam == 9)
        {
            ProcessKeyEvent(pUIMovie, uMsg, wParam, lParam);
            *pbNoFurtherProcessing = true;
        }
    }

    if (ASCII == 32) (ASCII == 9)
    {
        if (wParam == VK_SPACE || wParam == VK_TAB)
        {
            if (m_bToggleUI)
            {
                ToggleUI();
            }
            else
            {
                ToggleSettings();
            }
        }
    }
}

```

```

void GFXTutorial::ChangeMesh()
{
    if (m_bChangeMesh)
    {
        ChangeMesh();
    }
    else
    {
        ExternalInterface
        . ExternalInterfaceHandler::Callback / "A"
    }
}

```

```

static bool doChangeMesh = false;
static wchar_t changeMeshFilename[MAX_PATH] = L"";

...

if(strcmp(methodName, "MeshPath") == 0 && argCount == 1E

```


TextField.text
 Gfx::Movie::Invoke
 1 ("E 6.2.2).
 Scaleform
 > ' u - 3 u / "E Ũ

HUD > SetVariable
 / "A Q u □ ≠ ↑ □
 SetVariable "sticky" Gfx::Movie::SetVarType / "E ≠
 _root.mytextfield / "A 3 Y"E Ω
 r - SetVariable("_root.mytextfield.text", "testing", SV_Normal) Y"E 1 / "A
 3, ≠ ũ ' 1 ≠ . ≠ (> ≠) SV_Sticky " 3
 _root.mytextfield.text > / r ≠ , 3 / "A
 1 ≠ . ≠ C++ ≠ 1 1 ≠ SV_Normal ≠
 SV_Sticky / ≠ b 3 SV_Normal Q = ↓

SetVariableArray π — $/$ $\clubsuit$ $.$ $\neq$ $\spadesuit$
 — й $^{\circ}$ $\pounds$ $\neq$ $-$ — $/$ Q $\blacksquare$ $.$ π й
 GFxTutorial::InitGFx() / 1 $\sim$ $_root.SceneData$ й $>$ $\pounds$ й $.$

```
// Initialize the scene dropdown
Value sceneData[3];
sceneData[0].SetString("Scene with shadow");
sceneData[1].SetString("Show shadow volume");
sceneData[2].SetString("Shadow volume complexity");
pUIMovie->SetVariableArraySize("_root.SceneData", 3);
pUIMovie->SetVariableArray(Movie::SA_Value,
                           "_root.SceneData", 0, sceneData, 3);
```

Tutorial\Section6.1 / $\blacksquare$ й 1 $\spadesuit$ $\neq$ ExternalInterface $\emptyset$ $\sim$ $\blacksquare$ $\neq$
 w f $\pounds$ $\neq$, w f , й 3 ShadowVolume $\neq$ $\approx$ й

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```

        loadMovie unloadMovie
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```

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u , π r ^ "E ‡ f π bi' ... bi ' = . v
MovieClipLoader _ v Ÿ"E 1# .

```

```
var mclLoader:MovieClipLoader = new MovieClipLoader();
```

```

# ∅ / "A Q / □ "E # u "A
~ . f # _ v ~ # bi # .

```

```
// Load a file into a specific level
```

```

function LoadFlashLevel(url, level)
{
    trace("LoadFlashLevel(" + url + ", " + level + ")\n");
    mclLoader.loadClip(url, level);
}

```

```
//
```

```
// Load a file into a named object
```

```

function LoadFlash(url, objectName)
{
    trace("LoadFlashLevel(" + url + ", " + objectName + ")\n");
    var container:MovieClip = createEmptyMovieClip(objectName,
                                                    getNextHighestDepth());
    mclLoader.loadClip(url, container);
}

```

```

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# UI Ÿ □ # Ÿ ‡ ε # / ↑ ↑ ‡ ε # Æ / ↑ ↑ ` ♪
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```

```

    w
    ( ,_root.inventoryWindow.widget1.counter).

```

LoadMovie _root.myMovie / 5i ■ s £ — Ω , ‡
_root.counter π ‡ / ■ 1 ~ 3 u ‡ . ^ w ^ "E s
this:_lockout = true ẽ ʃ ‡ - . Lockroot π "E _root π ° ö
w "A s π 1 ‡ π 1 ‡ ̃ .
/ s / ° "E Q ‡ 3 "A ~ .

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LoadMovie ~ LoadMovieLevel ¶ √, ∫ π ßi ' . loadClip
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 MovieClipLoader π Ÿ Œ . Container fla ∅ π ‡
 √ · ↑ C++ ‡ ĭ ‡ π — ‡ ‡ √ Ÿ "E π
 placeholder √ √ .

41

```

//
// These callbacks are necessary because a movie does not
// necessarily load as soon as LoadMovie() returns.
//
// Currently the callbacks only print debug information.
// An application that needs to act on these events should
// either make ExternalInterface calls to notify the C++
// application or handle the events in ActionScript.

var mclListener:Object = new Object();

mclListener.onLoadError = function(target_mc:MovieClip, errorCode:String,
                                   status:Number)
{
    trace("Error loading image: " + errorCode + " [" + status + "]");
};

mclListener.onLoadStart = function(target_mc:MovieClip) : Void
{
    trace("onLoadStart: " + target_mc);
};

mclListener.onLoadProgress = function(target_mc:MovieClip,
                                     numBytesLoaded:Number,
                                     numBytesTotal:Number) : Void
{
    var numPercentLoaded:Number = numBytesLoaded / numBytesTotal * 100;
    trace("onLoadProgress: " + target_mc + " is " +
          numPercentLoaded + "% loaded");
};

mclListener.onLoadComplete = function(target_mc:MovieClip,
                                     status:Number) : Void
{
    trace("onLoadComplete: " + target_mc);
};

// Register the listener with the MovieClipLoader
mclLoader.addListener(mclListener);

```

The code in Tutorial\section6.2 /    d3d9guide.swf fxplayer.swf /
 container.swf r 1 / Ё . F8 Ё 3 HUD Ё ~ F9 Ё 3 r 
 UI Ё

```

void GFxTutorial::ProcessEvent(HWND hWnd, UINT uMsg,
                               WPARAM wParam, LPARAM lParam,
                               bool *pbNoFurtherProcessing)
{
    ...

    else if (wParam == VK_F8)
    {
        bool retval = pShellMovie->Invoke("_root.LoadFlashLevel",
                                           "%s, %d", "fxplayer.swf", 5);
    }
}

```


HUD F8 6.2 / 1 F5 1
 HUD . SetVariable _root.counter HUD _level5
 HUD / b , - .

```
void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam, LPARAM
                               lParam, bool *pbNoFurtherProcessing)
{
    int mx = LOWORD(lParam), my = HIWORD(lParam);

    if (pHUDMovie && uMsg == WM_KEYDOWN)
    {
        if (wParam == VK_F5)
        {
            int counter = (int)pHUDMovie->
                GetVariableDouble("_level5.counter");
            counter++;
            pHUDMovie->SetVariable("_level5.counter",
                Value((double)counter));
            char str[256];
            sprintf_s(str, "testing! counter = %d", counter);
            pHUDMovie->SetVariable("_level5.MessageText.text", str);
        }
    }

    ...
}
```

_level5.counter 6 d3d9guide.swf / , , +
 q + + / C++ ' + .

+ i , F8 HUD F5 1 . F2
 HUD F8 HUD F5 1 .
 HUD ... , 1 > + 0 1 - . + 1
 _level5 / , + . _level5.counter _global.counter - - .
 _global / , + / b ... b , / 1 .

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 Flash Studio / "A й SWF "E + SWF + / .
 + + 3 / + .

```

1.  MeshMaterialList
    Material {
        1.000000;1.000000;1.000000;1.000000;;
        40.000000;
        1.000000;1.000000;1.000000;;
        0.000000;0.000000;0.000000;;
        TextureFilename {
            "cellwallRT.jpg";
        }
    }

2.  DXUT
    A(parsing)
    DXUT
    cell.x

```


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 Gamebryo™ □ Unreal® Engine 3 □ ~

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y [Gfx::ImageCreator](#) ǎ ✚ Ç ☹

y [Gfx::Translator](#)